Umakant Sunil Kulkarni

Contact
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Current Address
Sunnyvale, CA 94086 USA

PROFILE SUMMARY

Ph.D. researcher specializing in network security, network protocols, and AI/ML for networked systems. Research and engineering experience spans protocol decoding, packet processing, zero-trust security for cloud-native networks, with a strong background in distributed cellular mobile core architectures and Wi-Fi Networks. First-author publications in leading ACM and IEEE venues, with a published AI-based systems patent in the United States, Germany, and China.

EDUCATION

Ph.D., Computer Science
Purdue University, West Lafayette, USA — Jan 2021 - Dec 2025
GPA 4.0

Master of Science, Computer Systems Networking
Northeastern University, Boston, USA — Sep 2015 - May 2017
GPA 3.71

Bachelor of Engineering, Electronics and Telecommunication
University of Pune, India — Aug 2011 - May 2015
First Class with Distinction

RESEARCH AND PROFESSIONAL EXPERIENCE

Palo Alto Networks, Santa Clara, CA USA

Senior Staff Researcher — Oct 2026 – Present

- Conduct research on network protocols, traffic analysis, and evasion techniques, focusing on protocol decoding for Advanced Threat Prevention in next-generation firewalls..
- Design and evaluate advanced security techniques for Frontier Virtual Patching, cloud-delivered threat protection, and agentic AI systems.

Nokia Bell Labs, Murray Hill, NJ USA

Research Scientist — Jan 2026 – Oct 2026

- Conducted research at the intersection of cellular networks (5G/6G), distributed systems, network security, and AI/ML to advance next-generation mobile network systems.
- Designed resilient distributed 6G core network architectures in collaboration with KDDI, focusing on core-network decomposition, distributed state management, fault tolerance, and service continuity.

Affirmed Networks/Microsoft, Acton MA USA

Principal Software Engineer — Jun 2017 – Dec 2020

- Developed stateless, containerized telecommunication software based on 5G standards for core network functions AMF, SMF, UPF, N3IWF, PCF and UDM that can be deployed via Docker and/or Kubernetes to achieve scalability, resiliency and redundancy.
- Implemented Istio service-mesh features into 5G NFs for observability, security and traffic management.
- Designed a build environment to define, install and upgrade applications on Kubernetes using Helm.
- Developed DPDK containerized UE-gNB simulator for testing different data-plane functionality.
- Built pyATS-enabled automation infrastructure and integrated it with JIRA and Jenkins for simulating CI/CD model of cloud-native microservice architecture.

PUBLICATIONS

- Modeling Latency of Split Autoregressive Decoding in Cellular Edge Systems
  Umakant Kulkarni, Kevin Di Lallo, Chonggang Wang, Robert Gazda, Sonia Fahmy
  Proc. of EI Workshop at ACM/IEEE Symposium on Edge Computing (SEC '26), 8pp., Oct 2026.

- Elastic and Energy-Efficient Management of Mobile Core Control Plane with In-Process WASM Workers
  Masayuki Kurata, Kyoshiro Muramatsu, **Umakant Kulkarni**, Van Quan Pham, Ahan Kak, Nakjung Choi, Masaki Suzuki
  IEEE International Conference on Network and Service Management (CNSM), 9 pp., Oct 2026

- Janus: A Dual-Mask Attention Transformer for Log-based Anomaly Detection in Cellular Networks
  Umakant Kulkarni, Sonia Fahmy
  Proc. ACM Meas. Anal. Comput. Syst. 10, 1, Article 14, (SIGMETRICS'26), 36pp., March 2026.

- RUN: A Case for Cross-Layer Networked Virtual Reality
  Yufeng Chen, **Umakant Kulkarni**, Voicu Popescu, Sonia Fahmy
  Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), 10 pp., October 2025.

- Maestro: QoE-Aware Dynamic Resource Allocation in Wi-Fi Networks
  Umakant Kulkarni, Khaled Diab, Lianjie Cao, Faraz Ahmed, Shivang Aggarwal, Puneet Sharma, Sonia Fahmy
  Proceedings of the ACM on Networking, Volume 3, Issue CoNEXT1, Article 4, 24 pp., March 2025

- Securing the Cloud-Native 5G Control Plane
  Umakant Kulkarni, Sonia Fahmy
  IEEE MILCOM Workshop on Secure FutureG Wireless Communications and Networked Systems, 6 pp., October 2024

- Understanding the Impact of Wi-Fi Configuration on Volumetric Video Streaming Applications
  Umakant Kulkarni, Khaled Diab, Shivang Aggarwal, Lianjie Cao, Faraz Ahmed, Puneet Sharma, Sonia Fahmy
  ACM SIGCOMM Workshop on Emerging Multimedia Systems (EMS), 6 pp., September 2023

- Toward QoE-based Routing Path Selection
  Umakant Kulkarni, Yufeng Chen, Patrick Melampy, Sonia Fahmy
  IEEE International Conference on High Performance Switching and Routing (HPSR), 6 pp., June 2023

- Slicure5G: Secure Slicing for 5G (Poster)
  Umakant Kulkarni and Sonia Fahmy
  USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2 pp. April 2023

- Towards A Low-Cost Stateless 5G Core (Poster)
  Umakant Kulkarni, Amit Sheoran, Sonia Fahmy
  IEEE International Symposium on Local and Metropolitan Area Networks, 2 pp., July 2022

- The Cost of Stateless Network Functions in 5G
  Umakant Kulkarni, Amit Sheoran, Sonia Fahmy
  ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS), 7 pp., December 2021

PATENTS

- QoE-aware Dynamic Resource Allocation
  Shivang Aggarwal, **Umakant Kulkarni**, Khaled Diab, Lianjie Cao, Faraz Ahmed, Puneet Sharma
  Published in USA, Germany and China, April 2026

FELLOWSHIPS AND AWARDS

- Elected to membership in The Honor Society of Phi Kappa Phi in January 2025. Membership is by invitation only and recognizes academic excellence among the top 10% of graduate students.
- Received "Employee Merit Recognition" award under the High-Profile Projects category for achieving significant milestones in "Securing 5G for Military Operations" research project at Purdue University
- 3rd Prize in Best-in-class 2022 Summer Interns Project Fair organized by Hewlett Packard Enterprise
- Received "Outstanding Graduate Teaching Assistant Award" for significant accomplishments and highest ratings in teaching and mentoring students at College of Engineering, Northeastern University.
- Awarded by KVPY Fellowship (English: Young Scientist Incentive Plan) with All India Rank – 5. This research fellowship was given by The IISc, Bangalore and Department of Science and Technology.
- Received Best Paper/Research/Poster award at seven regional conferences, India (2011-2015).

INTERNSHIPS

InterDigital, Conshohocken PA USA

Research Intern — May 2025 – Aug 2025

- Analyzed, designed, and implemented next generation 6G mobile networks.
- Developed 6G networking protocols, services, and solutions to enable application verticals (GenAI).

Hewlett Packard Enterprise, Milpitas CA USA

Research Associate Intern — May 2022 – Dec 2023

- Characterized QoE of volumetric video streaming over enterprise Wi-Fi networks, analyzing the impact of Wi-Fi control parameters and comparing QoE with traditional 2D video applications.
- Proposed an adaptive method for Wi-Fi resource allocation to maximize the overall system QoE and QoE fairness.

Affirmed Networks, Inc. Acton MA USA

SQA Intern — May 2016 – Jan 2017

- Laid the foundation for ePDG and T-WAG automation infrastructure; Built the 'smoke' as well as 'sanity' frameworks to automate the end-to-end workflow.
- Modified open-source libraries and configured StrongSwan module onto a linux box as well as Raspberry Pi to support different Wi-Fi calling features over IPsec.

Thuse Elektronics Pvt. Ltd., Pune India

Engineering Intern — Nov 2013 – Dec 2013

- Worked on the processes transformer manufacturing, PCB design, PCB assembling in line production, quality and rework management, process quality control, wave soldering on RoHS and non RoHS sections, automatic optical inspection and surface mount technology.

TEACHING EXPERIENCE

Purdue University, West Lafayette, IN USA Jan 2021 – Dec 2021

- Graduate Teaching Assistant for the course 'Computer Architecture (CS 250)'

Northeastern University, Boston, MA USA Jan 2017 – May 2017

- Graduate Teaching Assistant for the course 'Telecommunication Architecture and Systems (TELE 5320)' for 2 sections.

PROFESSIONAL SERVICE

- *ACM SIGCOMM 2026* — Artifact Evaluation Committee
- *IEEE INFOCOM 2027* — Technical Program Committee
- *PACT 2026* — Artifact Evaluation Committee
- *USENIX NSDI 2027* — Artifact Evaluation Committee
- *ACM CoNEXT 2026* — Artifact Evaluation Committee
- *ACM CAIS 2026* — Artifact Evaluation Committee
- *USENIX NSDI 2026* — Artifact Evaluation Committee
- *ACM SIGCOMM 2025* — Artifact Evaluation Committee
- *ACM IMC 2025* — Shadow Technical Program Committee
- **Journals:**
  - *IEEE Internet Computing*
  - *IEEE Transactions on Networking*
  - *IEEE Transactions on Network Science and Engineering*
  - *IEEE Transactions on Dependable and Secure Computing*

CERTIFICATIONS

Build Large Language Models (LLMs) from Scratch

Vizuara Technologies Private Limited · Issued Oct 2025 · Credential ID 0cd0775996b71249

Responsible Conduct of Research (RCR) Training - Faculty, Postdoctoral, and Graduate Students

CITI Program · Issued Oct 2025 · Expires Oct 2029 · Credential ID 66250359

Certified Telecommunication Engineer (Reg. No. – S13PUN128)

Bharat Sanchar Nigam Limited (BSNL), Govt. of India

- Platinum Level - Mobile Communication, IP Networking and Cyber Security Jul 2014
- Gold Level - Broadband Technology, Optical Fiber Technology Jan 2014
- Silver Level - Digital Switching and Transmission System, Telecom Support Infrastructure Jul 2013

WEB PROFILES

Homepage: `https://umakantkulkarni.github.io`

GitHub: `https://github.com/UmakantKulkarni/`

LinkedIn: `https://www.linkedin.com/in/umakantkulkarni/`

Google Scholar: `https://scholar.google.com/citations?hl=en&user=1v3d0A0AAAAJ`